

1. Introduction to Operations Management

Operations Management (OM) means managing everything needed to make products or deliver services efficiently.

It focuses on turning **inputs** (like materials, people, money, and information) into **outputs** (goods or services).

Why it's important:

Good operations mean lower cost, better quality, faster delivery, and more satisfied customers.

Main activities include:

- **Process design:** How work is done and in what order.
- **Capacity planning:** Making sure you have enough resources to meet demand.
- **Layout & facilities:** Arranging machines or desks to save time and reduce waste.
- **Workforce planning:** Managing staff schedules and training.
- **Procurement:** Buying the right materials at the right cost.
- **Quality control:** Ensuring products meet standards.
- **Performance measurement:** Tracking how efficiently things are done.

Process types:

- **Project:** One-time work (like construction).
- **Job shop:** Custom small batches.
- **Batch:** Medium volume, some variety.
- **Assembly line:** High volume, same products.
- **Continuous:** Runs 24/7 (like oil refining).

Key metrics:

Speed, productivity, delivery time, and defect rates.

Main idea:

Operations must balance cost, speed, and quality while keeping customers happy.

2. Supply Chain Management (SCM)

SCM means managing the whole journey of a product — from suppliers to customers.

It connects **suppliers, manufacturers, distributors, and retailers** to deliver goods efficiently.

Why it matters:

It reduces cost, improves delivery speed, and prevents shortages.

Main parts:

- **Sourcing:** Choosing and managing suppliers.
- **Production:** Making the goods.
- **Logistics:** Moving and storing goods.
- **Forecasting:** Predicting customer demand.
- **Inventory:** Balancing stock levels.
- **Returns:** Handling returned or damaged items.
- **Information systems:** Using software (like ERP, TMS, WMS) for tracking and planning.

Common problems:

- **Bullwhip effect:** Small demand changes cause big problems in the chain.
- **Single supplier risk:** Depending on one supplier can cause issues.
- **Long lead times:** Slow delivery from suppliers.

Modern tools:

Use data, technology, and planning systems for real-time updates and sustainability (like green supply chains).

3. Inventory Management

Inventory management controls how much stock you buy, store, and sell — to meet demand without wasting money.

Types of inventory:

- **Raw materials:** Basic inputs.
- **Work-in-process:** Items being made.
- **Finished goods:** Ready for customers.
- **Safety stock:** Backup supply.
- **In-transit:** Goods being shipped.

Costs involved:

- **Holding costs:** Storing items.
- **Ordering costs:** Cost per purchase.
- **Stockout costs:** Losing sales if items are unavailable.

Control methods:

- **Continuous review:** Keep checking and reorder when low.
- **Periodic review:** Check at fixed times.
- **ABC analysis:** Focus more on costly items (A items).
- **JIT (Just-In-Time):** Keep very low inventory, order only when needed.

Key measures:

Inventory turnover, days of inventory, stockout rate.

Main goal:

Keep the right amount of stock — not too much, not too little.

4. Lean Manufacturing

Lean is about making things with **less waste** and **more value**.

It started with Toyota and focuses on **continuous improvement**.

Five main ideas:

1. Understand what the customer values.
2. Identify and remove waste.
3. Make work flow smoothly.
4. Produce only what is needed (pull system).
5. Keep improving all the time.

8 types of waste (TIMWOODS):

Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, Skills underuse.

Popular tools:

- **5S:** Keep workplace clean and organized.
- **Kaizen:** Small daily improvements.
- **Kanban:** Card system to control production.
- **Poka-Yoke:** Mistake-proofing.
- **TPM:** Maintain machines to avoid breakdowns.

Result:

Faster delivery, better quality, less waste, happier customers.

5. Operations Planning and Scheduling

This means deciding **what to produce, when, and how**, so resources are used wisely.

Levels of planning:

- **Strategic:** Long-term goals (like new factories).

- **Tactical (Aggregate):** Medium-term balance of demand and resources.
- **Operational:** Short-term daily or weekly plans.
- **Scheduling:** Assigning specific tasks to workers or machines.

Main tools:

- **MRP:** Calculates materials needed based on product plans.
- **ERP:** Combines all departments into one system.
- **APS:** Advanced software for detailed scheduling.
- **S&OP:** Aligns sales and production plans.

Key measures:

On-time delivery, lead time, machine use, order backlog.

Purpose:

Ensure materials, people, and machines work together to meet customer needs efficiently.

6. Quality Management and Continuous Improvement

This ensures that products and services meet standards and keep getting better over time.

Main approaches:

- **TQM:** Everyone is responsible for quality.
- **Six Sigma:** Use data to reduce defects.
- **PDCA cycle:** Plan → Do → Check → Act.
- **ISO 9001:** International quality standard.

Tools used:

Control charts, root cause analysis (5 Whys, Fishbone diagram), process capability studies.

Costs of quality:

- **Prevention:** Training and design.
- **Appraisal:** Testing and inspection.
- **Internal failure:** Scrap or rework.
- **External failure:** Complaints, returns.

Main idea:

Don't just check for defects — prevent them and make improvements regularly.

7. Logistics and Distribution Management

This handles **how products move** from the factory to the customer.

Main parts:

- **Transportation:** Choosing the best mode (air, sea, road, rail).
- **Warehousing:** Storing and managing goods.
- **Order fulfillment:** Picking, packing, and shipping products.
- **Network design:** Deciding where to place warehouses.
- **Reverse logistics:** Managing returns or recycling.

Technology helps:

- **WMS (Warehouse Management System)** for tracking inventory.
- **TMS (Transportation Management System)** for planning routes.
- **Automation & tracking tools** make deliveries faster and cheaper.

Measures:

On-time delivery, shipping cost, order accuracy, and delivery time.

Goal:

Deliver the right product, to the right place, at the right time, and at the lowest cost.